

Automating Acoustic and Articulatory Segmentation Exercises

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Background: Phonetic Segmentation

- ▶ Segmenting audio data has been part of our phonetic analysis toolset since the invention of audio recordings.
- ▶ Since then we have gained two large improvements:
 - ▷ spectrograms (Gabor 1946, Koenig et al. 1946, Fant 1961) and
 - ▷ computer aided segmentation with programs such as Praat (Boersma and Weenink 2010).
- ▶ Segmentation has become more complex with the introduction of articulatory recording methods and other multimodal data, and the need to analyse them in conjunction with traditional audio recordings.
- ▶ Now we are on the cusp of yet another great shift in how we work with the rapid improvement of forced alignment systems which started already before the current revolution in deep learning (see e.g. Rosenfelder et al. 2022) and is now becoming part of that revolution (McAuliffe et al. 2017, Kelley et al. 2024).

Purpose: Teach Segmentation with Minimal Effort

- ▶ On the surface it might seem that the days of us needing to do manual segmentation are coming to an end, but we still need expert human segmenters for two crucial tasks:
 1. The deep learning frameworks need data to learn from and this data needs to be well matched with what ever dataset we wish to eventually analyse.
 2. Even if it should happen that we get a magical large segmentation model that can deal with any human language and any dialect and accent there of – even ones that do not exist yet – we will still need to verify the output of any such model.
- ▶ In addition we are getting more and more time domain articulatory data.
- ▶ Learning to segment data helps students develop an understanding of the concepts of phoneme and phone.
- ▶ Yet, surprisingly, we do not have easy methods for teaching segmentation to new learners – neither for generating exercises nor for running them.

Method: Automated Exercises

- ▶ A potential solution for this problem is generating segmentation exercises from existing data:
 - ▷ Take any Praat TextGrids and the corresponding data – whether audio only or multimodal – and scramble the boundary placement within the TextGrid.
 - ▷ Then let the learner correct the boundaries and compare to the original as they go along.
- ▶ This all has been implemented in PATKIT – Phonetic Analysis ToolKIT (Palo et al. 2026).
- ▶ PATKIT has been written in the Python programming language.
- ▶ It can also process articulatory data.
- ▶ Get PATKIT on PyPi: <https://pypi.org/project/patkit/> or fork it on GitHub: <https://github.com/giuthas/patkit>.

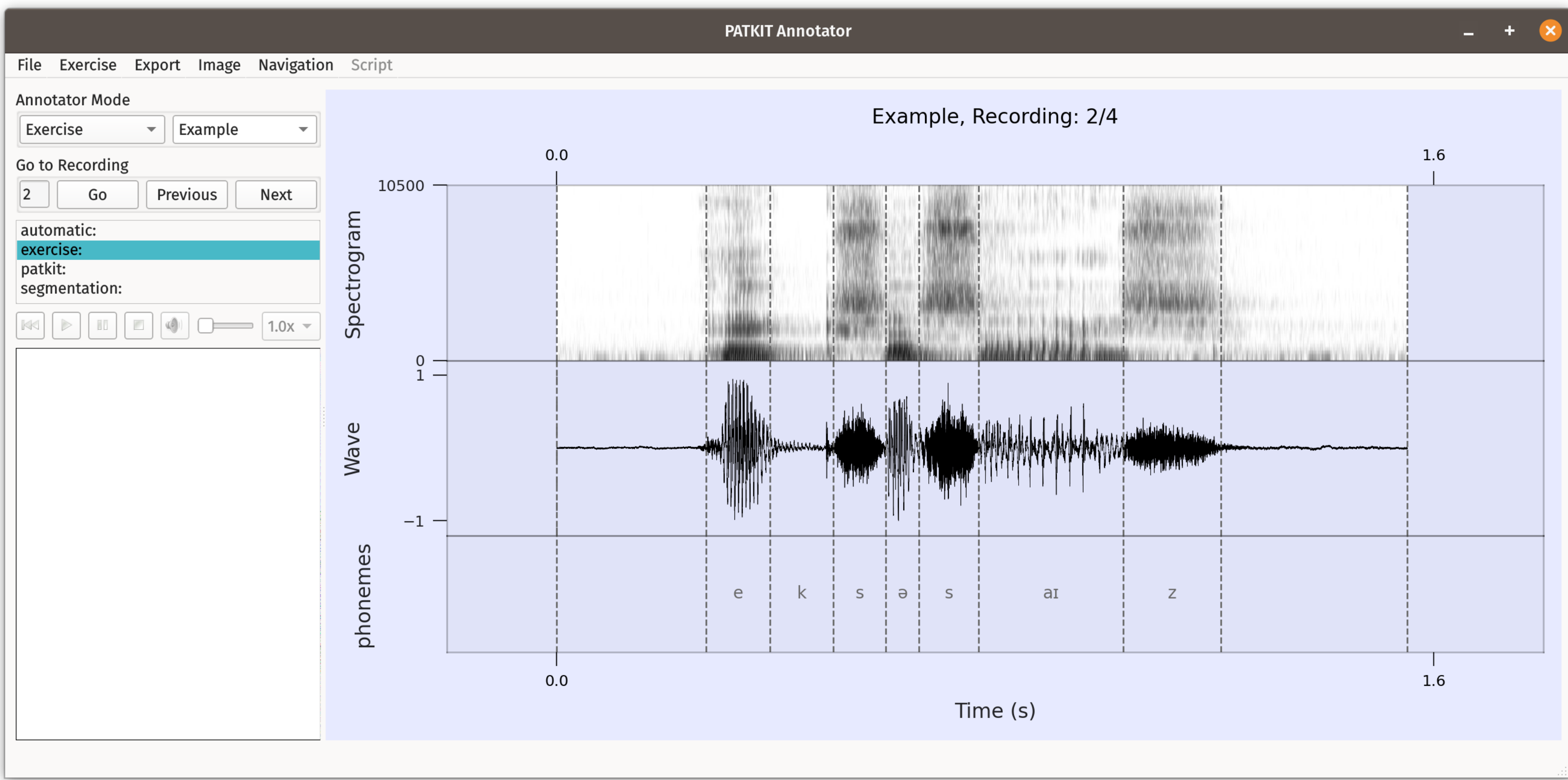
Discussion: Future work

- ▶ I would like to know what you need.
- ▶ If you wish to contribute ideas, feature requests, bug reports, documentation, more exercises, code, or something else it would all be welcome.

References

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User interface: Example Answer



User interface: Scrambled TextGrid

